
RENAL
FAILURE
HIGH
CREATININE
190
110

Acute renal failure (ARF) is a sudden and almost complete loss of kidney function (decreased GFR) over a period of hours to days.

Categories of Acute Renal Failure

Pre renal (hypo perfusion of kidney 55-60%),

Intra renal (actual damage to kidney tissue 5%)

Post renal (obstruction to urine flow 35-40%).

Pre-renal conditions occur as a result of impaired blood flow that leads to hypo perfusion of the kidney and a drop in the GFR. Common clinical situations are impaired cardiac performance (myocardial infarction, or heart failure), and vasodilation (sepsis).

Intra-renal causes of ARF: are the result of actual parenchymal damage to the glomeruli or kidney tubules. Conditions such as burns crush injuries, and infections, as well as nephrotoxic agents, may lead to acute tubular necrosis and cessation of renal function.

- **Post-renal causes of ARF: are usually the result of an obstruction somewhere distal to the kidney. Pressure rises in the kidney tubules; eventually, the GFR decreases. Although the exact pathogenesis of ARF and oliguria is not always known, The following conditions that reduce blood flow to the kidney and impair kidney function:(1)hypovolemia; (2) hypotension; (3) reduced cardiac output; (4) obstruction of the kidney or lower urinary tract by tumor, or kidney stone; and (5) bilateral obstruction of the renal arteries or veins.**


Common Causes of
Acute Renal Failure (ARF)
Pre-Renal Failure
Hemorrhage
DIURETICS
Renal Losses (Diuretics)
GI Losses (Vomiting, Diarrhea)
Myocardial Infarction
Heart Failure
Sepsis
BP MEDS
Anaphylaxis
Antihypertensive Medications
القشل اكلوي ← تلف الاعءاء ← تراكم السموم


Common Causes of
Acute Renal Failure (ARF)
Intra-Renal Failure
Hemorrhage
Myoglobinuria (Trauma, Burns)
Renal Losses (Diuretics Reaction)
GI Losses (Vomiting, Diarrhea)
Hemoglobinuria
(Transfusion Reaction)
ANTIBIOTICS
Gentamicin
Antibiotics
(Gentamycin)
Heavy Metals
(Lead, Mercury)
NSAIDs
Anaphylaxis
Acute
Pyelonephritis
الفشل الكلوي → تلف الأعضاء → تراكم السموم
Common Causes of
Acute Renal Failure (ARF)
Post-Renal Failure
Kidney Stones (Calculi)
Tumors & Strictures
ANTIBIOTICS
Gentamicin
Antibiotics
(Gentamycin)
Heavy Metals
(Lead; Mercury)
NSAIDS
NSAIDs
(Non-Steroidal Anti-Inflammatories)
Acute Pyelonephritis
الفشل الكلوي → تلف الأعضاء → تراكم السموم

Phases of Acute Renal Failure

The initiation period begins with the initial insult and ends when oliguria develops

The oliguria period is accompanied by a rise in the serum concentration of substances usually excreted by the kidneys (urea, creatinine, uric acid

In the diuresis period, the third phase, the patient experiences gradually increasing urine output, which signals that glomerular filtration has started to recover.

The recovery period signals the improvement of renal function and may take 3 to 12 months. Laboratory values return to the patient's normal level.

Assessment and Diagnostic Findings

Changes in Urine

Urine output varies (scanty to normal volume), hematuria may be present, and the urine has a low specific gravity (1.010 or less).

Change in Kidney Contour

Ultrasonography is a critical component of the evaluation of both acute and chronic renal failure.

The BUN level rises steadily at a rate dependent on the degree of catabolism (breakdown of protein), and protein intake.

Hyperkalemia

With a decline in the GFR, the patient cannot excrete potassium normally. Patients with oliguria and anuria are at greater risk for hyperkalemia than those without oliguria.

Metabolic Acidosis

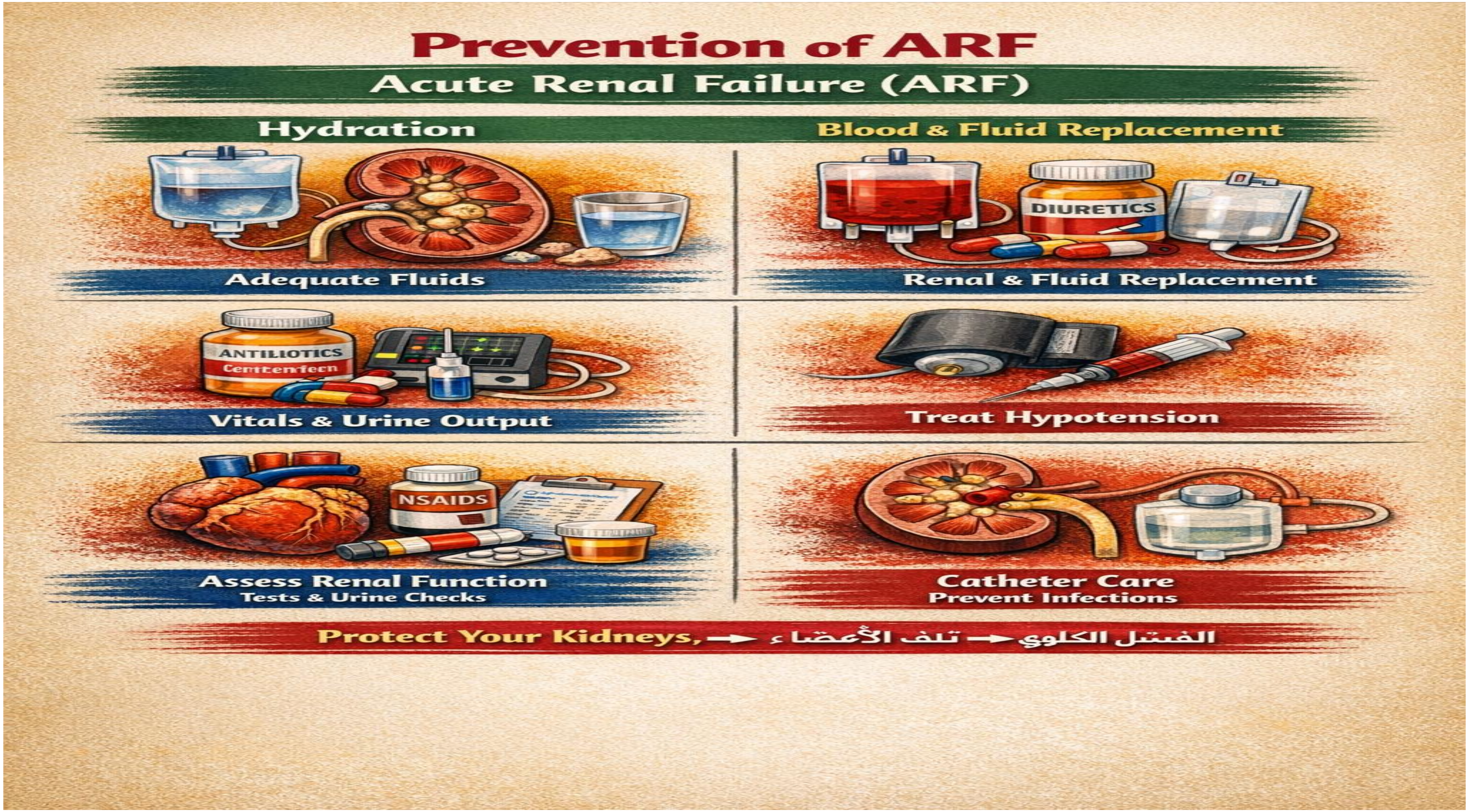
Patients with acute oliguria cannot eliminate the daily metabolic load of acid-type substances produced by the normal metabolic processes.

Calcium and Phosphorus Abnormalities

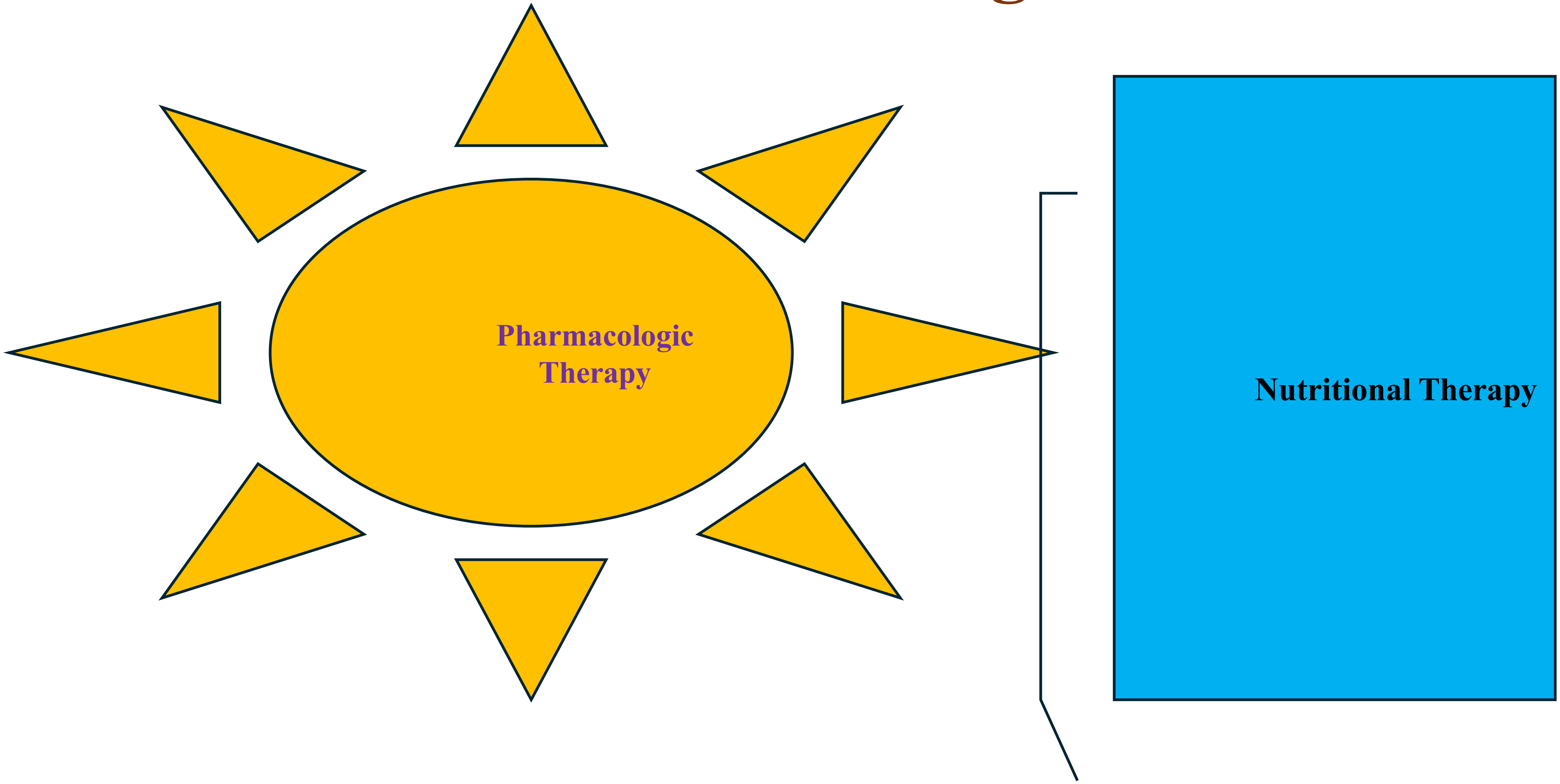
There may be an increase in serum phosphate concentrations; serum calcium levels may be low in response to decreased absorption of calcium from the intestine and as a compensatory mechanism for the elevated serum phosphate levels.

Anemia

Anemia inevitably accompanies ARF due to reduced erythropoietin production, uremic GI lesions, reduced RBC life span, and blood loss, usually from the GI tract.


Prevention of ARF
Acute Renal Failure (ARF)
Hydration
Blood & Fluid Replacement
Adequate Fluids
DIURETICS
Renal & Fluid Replacement
ANTILIOTICS
Vitals & Urine Output
Treat Hypotension
NSAIDS
Assess Renal Function
Tests & Urine Checks
Catheter Care
Prevent Infections
Protect Your Kidneys, → تلف الأعضاء → الفشل الكلوي

Medical Management




Are there any questions?